ACRA

Adaptive Color Re-Encoding Algorithm with CNN-Based Semantic Analysis for Daltonism-Aware Enhancement of Public Visual Materials

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Background of the Study

The Core Problem

- Public color-coded materials — signs, evacuation maps, safety posters — rely on hue alone to communicate critical information
- When colors lack sufficient luminance contrast, individuals with CVD experience perceptual collapse
- Outdoor glare worsens this — even well-designed signs become unreadable

Who Is Affected?

- ~300 million people globally have some form of CVD
- Red-green deficiencies account for 95% of all CVD cases
- Deuteranomaly is the most prevalent subtype — ~6% of males worldwide
- In the Philippines: ~3 million Filipinos affected; no national screening program exists

The Gap: Existing tools (OS high-contrast modes, Daltonize) apply uniform transformations that destroy visual cues — none provide semantic-aware, targeted re-encoding that preserves the original design intent.

Project Context

ACRA operates within the Philippine regulatory landscape — where color-coded systems govern public safety across sectors:

Healthcare (DOH)

Color-coded waste bins: Black, Green, Yellow, Orange, Red
Medical infographics, wayfinding posters, emergency markers

Disaster Preparedness (PAGASA)

Tri-color rainfall alerts: Yellow (awareness) → Orange (preparation) → Red (emergency)
Deutan users struggle to distinguish these transitions

Transportation (DPWH & Metro Rail)

Traffic & road signs, railway maps, pedestrian navigation
Environmental deterioration reduces contrast further

Local Gov't & Public Service

Fire exit diagrams, safety infographics, public announcements
Color encoding falls within Deutan confusion spectral zone

Purpose & Description

ACRA improves visual accessibility for CVD individuals by integrating a CNN-based object detector with a cloud-based adaptive re-encoding framework — making color-critical public materials distinguishable without altering their original design intent.

4 Key System Functions

01 · IoT Capture

Raspberry Pi 5 + Camera Module 3
Captures real-world public materials
Auto-uploads to cloud

02 · Semantic Detection

YOLOv8m detects color-critical ROIs
Filters out non-informational regions
Targeted, not global

03 · ACRA Re-encoding

CVD simulation (Machado 2009)
CIEDE2000 & CIE76 metrics
LCH color optimization

04 · Evaluation & Output

CCCM: ΔE improvement
Conflict resolution rate
Naturalness preservation

Statement of the Problem

How can ACRA provide a scalable solution for improving color interpretability in public communications?

SOP 1	How is ACRA conceptualized and planned as a system for Daltonism-aware enhancement of public visual materials?	**Planning**
SOP 2	How do researchers design and develop a system to detect and re-encode color conflicts in public visual materials?	**Development**
SOP 3	How is ACRA evaluated using ISO/IEC 25010 across three respondent groups?	**Evaluation**

SOP 3 sub-groups: IT Experts (Functional Suitability, Performance Efficiency, Usability, Reliability) · Optometrists (Functional Suitability, Performance Efficiency, Usability) · Deutan CVD Respondents (Functional Suitability, Usability)

Objectives of the Study

Objective 1 — Conceptualize & Plan

Define scope, architecture, tools, and hardware; produce the full system design blueprint for ACRA with YOLOv8m + cloud backend.

Objective 2 — Design & Develop

Build the system: CNN detection stage (YOLOv8m), ACRA re-encoding framework (CIEDE2000/CIE76/LCH), FastAPI backend, React frontend, IoT integration.

Objective 3 — Evaluate via ISO/IEC 25010

Assess ACRA using a standardized quality model with three respondent groups across Functional Suitability, Performance Efficiency, Usability, and Reliability.

Scope & Limitations

SCOPE

- Targets Deuteranomaly & Deuteranopia specifically
- Input: single-frame images (JPEG, PNG, WebP, BMP, TIFF, GIF, AVIF, HEIC)
- Two input paths: Raspberry Pi IoT auto-upload + manual browser upload
- Bulk uploads: up to 50 images, 10 MB each
- Processing: server-side FastAPI on Render
- Storage: Supabase (24-hour auto-delete for privacy)
- Frontend: React on Vercel — desktop browser only
- CVD severity slider: anomalous trichromacy → complete dichromacy

LIMITATIONS

- Not a medical treatment — defined as a digital visual aid only
- No video support — static frame processing only
- Server latency is network-dependent
- Free-tier Supabase constraints limit storage & API rate
- No mobile web app; no in-browser camera capture
- PDFs and non-image formats are unsupported
- Does not address Protanopia/Achromatopsia
- Algorithmic fidelity bounded by Machado (2009) mathematical model

Review of Related Literature — 5 Key Studies

1 **Machado, Oliveira & Fernandes (2009) — Physiologically-Based CVD Simulation**

Introduced a continuous spectral-shift simulation model for anomalous trichromacy and dichromacy. ACRA directly uses this as its CVD simulation engine.

2 **Jocher et al. (2023) — Ultralytics — YOLOv8 Architecture**

Established YOLOv8m as a state-of-the-art single-stage detector. ACRA uses YOLOv8m for semantic region localization — the front-end of the re-encoding pipeline.

3 **Sharma, Wu & Dalal (2005) — CIEDE2000 Color Difference Formula**

Defined CIEDE2000 as the perceptually uniform color-difference standard. ACRA uses ΔE_{00} as both its conflict detector and re-encoding optimization objective.

4 **Jamil & Denes (2024) — MDPI Computers — Limitations of OS High-Contrast Modes**

Demonstrated that universal high-contrast modes destroy meaningful visual cues rather than enhancing them — directly motivating ACRA's semantic-priority approach.

5 **Tapang & Espineda (2024) — Filipino CVD Lived Experiences in Metro Manila**

Documented that Filipino CVD individuals face daily accessibility barriers in Metro Manila environments — the only local study of its kind, justifying ACRA's Philippine focus.

Research Method

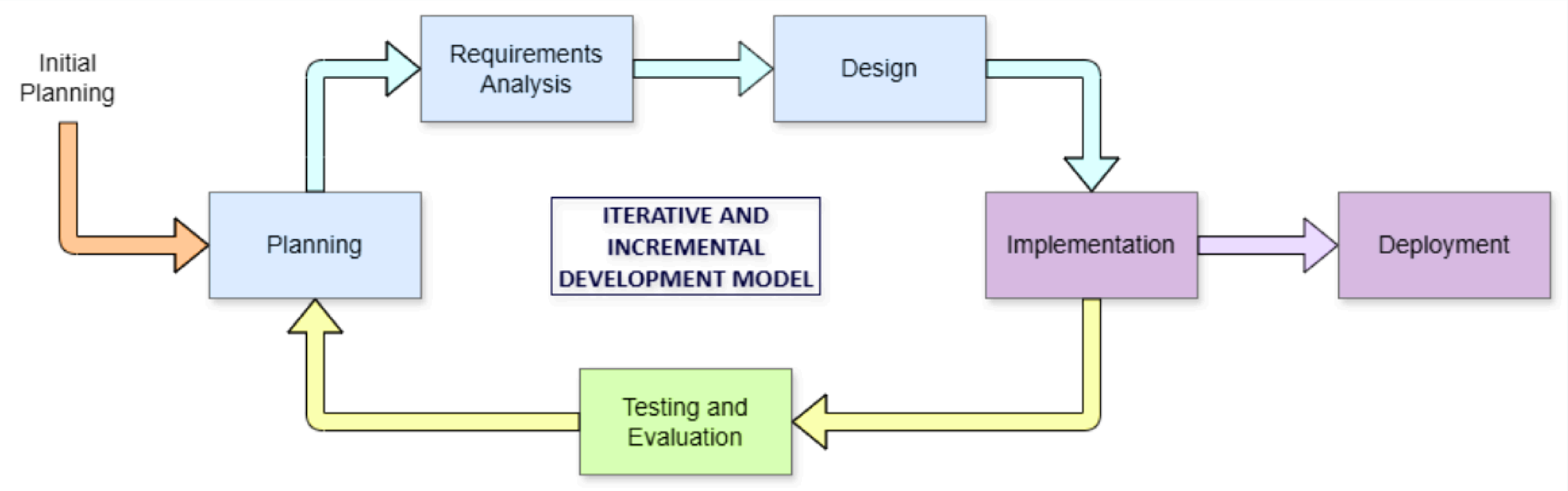
Applied Developmental Research

Goal: Build & assess a fully operational computational system — not theoretical outputs. Performance measured via objective mathematical metrics (ΔE, mAP50, conflict resolution rate, latency).

Quantitative + Qualitative

Quantitative: CNN performance metrics + re-encoding color-difference scores. Qualitative: User-based system assessment via ISO/IEC 25010 Likert survey across 3 respondent groups.

Iterative & Incremental SDLC (Agile-aligned)



Feedback loop from Testing → Planning drives each iteration

Requirements Analysis

Functional Requirements

- Detect & localize color-critical ROIs in arbitrary uploaded images
- Simulate CVD perception at user-specified severity (Machado model)
- Identify color pairs that become indistinguishable under simulation
- Correct those pairs while preserving visual naturalness for normal viewers
- Accessible from any browser — no client installation required

Non-Functional Requirements

Server Latency	< 5 seconds (full pipeline)
mAP50 (CNN)	Pass ≥ 0.60 | Target ≥ 0.70
ΔE Improvement	> 15 per conflict pair
Conflict Resolution	> 80% of flagged pairs
Naturalness Preservation	ΔEoriginal < 12
Data Privacy	Row-Level Security + 24h auto-delete

Training Dataset Summary

Pre-Augmentation:	**1,274 images**	Annotations:	**33,774 boxes**
Post-Augmentation:	**3,312 images**	Train/Val/Test Split:	**80% / 10% / 10%**

System Architecture — Design Phase

End-to-End Cloud Architecture

IoT Layer

Raspberry Pi 5
+
Camera Module 3

Capture → Auto-Upload
via picamera2

REST API

Cloud Backend (Render)

FastAPI

↓ ONNX Runtime
YOLOv8m Inference

↓ ACRA Framework
CVD Simulation
FCM Clustering
Conflict Detection
Color Re-encoding

↓ Metrics (ΔE, CRR)

HTTPS

Frontend (Vercel)

React Web UI
Upload / CVD Params
Results Dashboard
ΔE Metrics Display

Database (Supabase)

PostgreSQL
Supabase Auth (JWT)
Image Storage
24h Auto-Delete

Theoretical Framework

1 **Trichromatic Theory of Color Vision — Young (1802) / Helmholtz**

Explains the physiological basis of CVD — missing or shifted M-cones (Deutan). Informs why specific color pairs collapse perceptually.

2 **Machado CVD Simulation Model — Machado, Oliveira & Fernandes (2009)**

Physiologically-based spectral shift model that unifies anomalous trichromacy & dichromacy. ACRA's core simulation engine.

3 **YOLOv8m Single-Stage Detection — Jocher et al. (2023) — Ultralytics**

CSPDarknet backbone + C2f FPN neck + detection head. Provides semantic region localization (ROI) in a single forward pass.

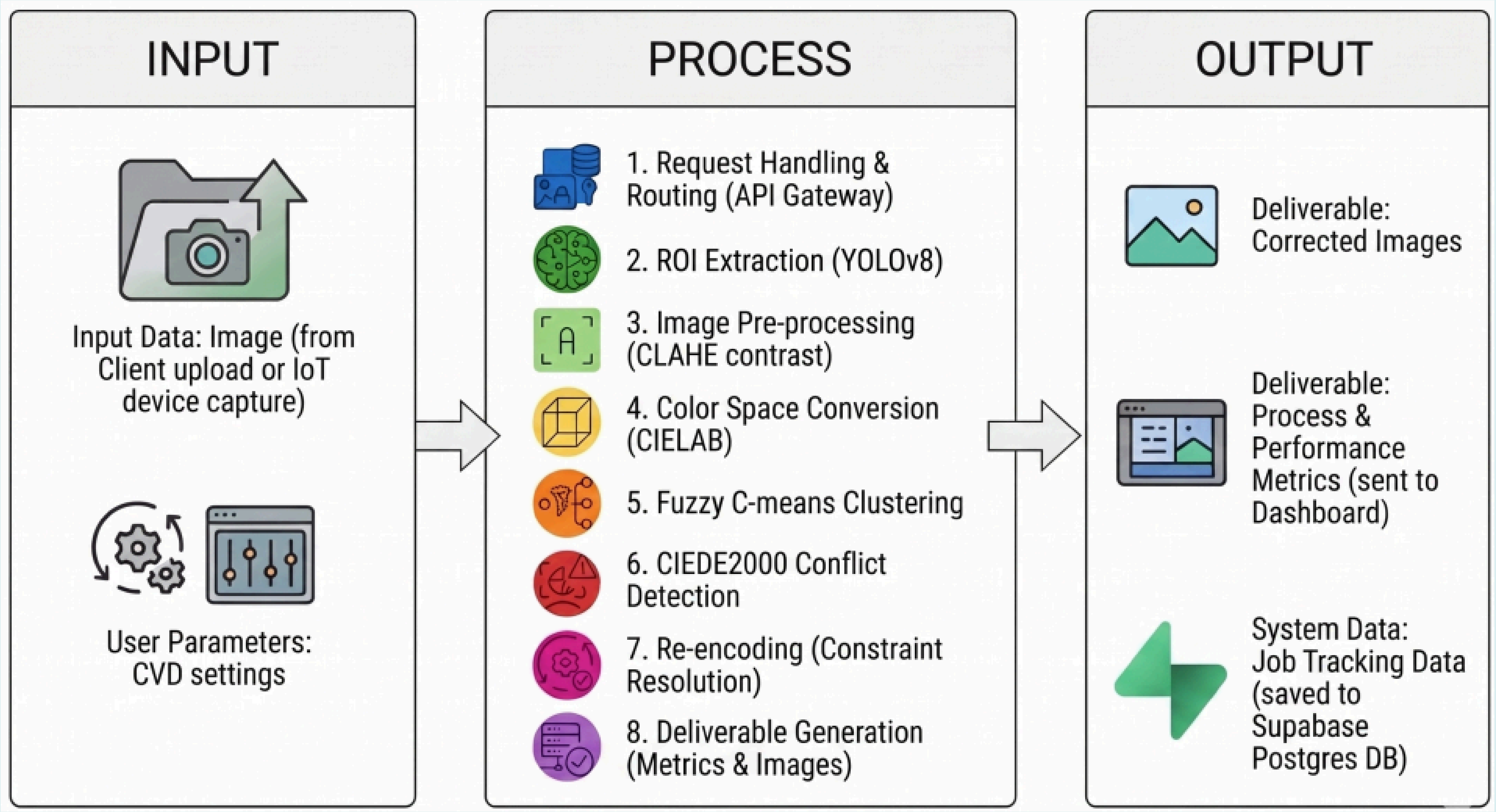
4 **CIEDE2000 (Sharma et al., 2005) — International standard ΔE_{00}**

Perceptually uniform color-difference metric. Used as ACRA's conflict detector, optimization objective, and evaluation measure.

5 **Fuzzy C-Means Clustering — Bezdek (1981)**

Soft cluster membership in CIELAB space enables smooth per-pixel reconstruction of re-encoded colors via inverse-distance weighting.

Conceptual Framework — ACRA End-to-End Pipeline



Implementation

FastAPI Backend

Image validation, JWT auth, ONNX Runtime inference session (loaded once at startup), ACRA algorithm execution, metrics computation, Supabase job records (JSONB)

YOLOv8m (ONNX)

Exported opset-12 ONNX from best.pt checkpoint. ROIDetector class: resize→640, normalize, NMS@IoU=0.50. Returns clean pixel-coordinate bounding boxes

ACRA Re-encoder

Python: CLAHE→linear RGB→LMS→CIELAB→FCM→conflict detection→bounded optimization→pixel reconstruction→sRGB output + metrics dict

React Frontend (Vercel)

Image upload, CVD parameter selection, bounding box overlay, 4-state view (original/simulated/corrected/sim-corrected), JSON/CSV metric export

Supabase Integration

Auth: JWT + Google OAuth; Storage: user-scoped paths; PostgreSQL: JSONB job records; RLS: per-user isolation; Edge Function: 24h auto-delete cron

Raspberry Pi IoT

picamera2 library, researcher-initiated capture at 3840×2160, timestamped filenames, background auto-upload script → FastAPI or Supabase Storage

Deployment

Render	**FastAPI Backend** Manages uploads, ONNX inference, ACRA re-encoding, metric computation, HTTPS API endpoints
Vercel	**React Frontend** Browser-accessible UI — no client install. Desktop-only. Image upload, CVD params, results dashboard.
Supabase	**Auth + Database + Storage** JWT auth (+ Google OAuth), PostgreSQL (JSONB job records), Row Level Security, Edge Function 24h auto-delete
Raspberry Pi 5	**IoT Edge Device** Camera Module 3 capture + auto-upload script. No CNN inference on device. Data acquisition only.

Note: As of May 2026, full cloud deployment to Render/Supabase/Vercel will proceed immediately after final integration testing is complete.

Data Collection & Preparation

Multi-source strategy for authentic real-world variation:

Primary: Raspberry Pi 5

Field capture — roads, buildings, institutions in Quezon City. Intentionally varied lighting, glare, angles. No staging.

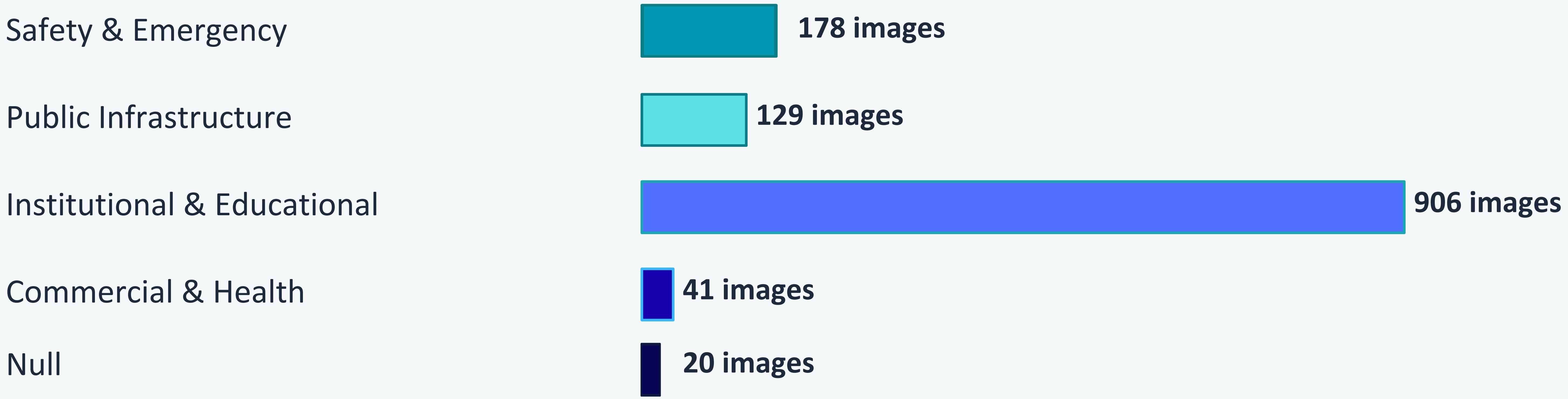
Softcopy: Partner Institutions

OLFU, Quezon City government, Barangay offices — brochures, public info displays, educational graphics.

Supplemental: Smartphone

iPhone 13 when Pi wasn't deployed. Same subject categories. Adds sensor-level diversity (color rendering, compression).

Dataset by Material Category (Pre-Augmentation)



Total pre-augmentation: 1,274 images • Post-augmentation: 3,312 images • 33,774 bounding box annotations

Data Annotation

5-Class annotation schema using Roboflow — each class maps directly to a downstream re-encoding function:

Class	Description	Count	Share
roi-object	**Full outer boundary of a sign, label, or packaging unit** *Exclude: Backgrounds; surfaces fully covered by more specific boxes*	**1,929**	5.7%
roi-text	**All text, numbers, alphanumeric content visible in frame** *Exclude: Completely unreadable text (100% glare/darkness)*	**22,241**	65.9%
roi-color	**Regions where color itself is the entire message (panels, bands)** *Exclude: Incidental/decorative colored surfaces with no communicative function*	**2,726**	8.1%
roi-symbol	**Icons, pictograms, hazard triangles, government seals, symbols** *Exclude: Logos where color is purely decorative*	**6,183**	18.3%
exclude-person	**Any real human visible in frame (full/partial body, face, hand)** *Exclude: Cartoon figures on signs (label as roi-symbol instead)*	**695**	2.1%

CNN Model Development — YOLOv8m

Architecture

Backbone (CSPDarknet)	Neck (C2f + FPN)	Detection Head
Hierarchical feature extraction across multiple spatial scales. Pretrained on COCO 80-class dataset. → Learns edges, textures, shapes, semantic structures	Combines multi-scale feature maps → consistent detection for both large & small ROIs in the same frame. → Critical for close-up signs vs. distant signage	Predicts bounding boxes & confidence scores across 8,400 anchor points on 640×640 inference grid. → Retrained from scratch for ACRA's 5-class task

Two-Stage Training Process

Parameter	Stage 1 — Frozen Backbone	Stage 2 — Full Fine-tuning
Starting Weights	yolov8m.pt (COCO pretrained)	Stage 1 best.pt
Frozen Layers	Backbone (first 10 layers)	None — entire network trains
Epochs / LR	25 epochs / lr=0.01	30 epochs / lr=0.001
Purpose	Train detection head only (protects backbone features)	Refine all parameters toward ACRA's specific visual domain

→ Trained on Kaggle (NVIDIA T4 GPU, 16GB VRAM) • Exported to ONNX format (opset 12) • Loaded once at FastAPI startup

Adaptive Color Re-Encoding Algorithm (ACRA)

Novel contribution: semantic-aware, perceptually-bounded color re-encoding for Deutan CVD accessibility

PREPROCESSING	CLAHE on LAB L-channel → inverse sRGB gamma (γ=2.4) → linear RGB → XYZ/D65 → CIELAB + Machado LMS matrix (severity-interpolated) for CVD simulation in parallel
SEGMENTATION	Fuzzy C-Means (m=2.0, k-means++ init): k=3 for ROI<32px, k=5 otherwise; max 12,000px subsample Outputs: soft cluster centers in CIELAB space
CONFLICT DETECTION	Two-pass hybrid: CIE76 rapid pre-check → CIEDE2000 validation (threshold 20; elevated to 35 for red-green pairs) Pin near-neutral clusters; discard near-duplicates (ΔE<12)
CONSTRAINED RE-ENCODING	3-pass optimization: (1) conservative L+C shifts, (2) larger naturalness budget, (3) hue-only for unresolved pairs Goal: $\Delta E_{sim} \geq 24$ between conflict pairs, $\Delta E_{original} \leq 12$ for normal-vision viewers
PIXEL RECONSTRUCTION	Inverse-distance weighting of cluster center shifts → smooth spatial transitions → convert back to sRGB uint8

System Integration

Fixed interface contract between detection and re-encoding stages enables modular development and isolated validation:

POST /process (image + CVD params)	Auth (JWT) MIME + size validation	YOLOv8m ONNX Detect ROIs → bounding boxes	ACRA Framework Sim + Cluster + Re-encode	Store to Supabase + Update job record (JSONB)

Auto-Upload Script (Raspberry Pi 5)

- After capture: timestamped image saved to SD card
- Background Python script posts image to FastAPI endpoint or Supabase Storage
- Handles direct upload when connected; queues for next opportunity otherwise
- Pi's role ends at transmission — zero inference or processing on device

RealVNC — development-only remote access tool; no presence in deployed system

Testing & Validation Framework

Evaluated against ISO/IEC 25010 + computational performance metrics + hardware thermal validation:

Functional Suitability

Metric: mAP50, ΔE Improvement per ROI, Conflict Resolution Rate

Respondents: IT Experts + Optometrists + Deutan CVD

Performance Efficiency

Metric: Server inference time (<2s), full framework latency (<5s)

Respondents: IT Experts + Optometrists

Usability

Metric: 5-point Likert survey — interface clarity, navigation, output readability

Respondents: All three respondent groups

Reliability

Metric: Repeated runs on same test images — pass/fail consistency

Respondents: IT Experts

Hardware Thermal Validation: Pi 5 under continuous 30-min capture session — must stay below 80°C (firmware throttle threshold)

Statistical Treatment of Data

Population & Sampling

IT Experts (n=5) — Purposive Sampling

Evaluates: Functional Suitability, Performance Efficiency, Usability, Reliability

Licensed Optometrists (n=5) — Purposive Sampling

Evaluates: Functional Suitability, Performance Efficiency, Usability

Deutan CVD Respondents (n=10) — Purposive Sampling

Evaluates: Functional Suitability, Usability

Rationale: 5 experts per group is sufficient per Nielsen & Landauer (1993) — detects ~85% of quality issues. 10 Deutan CVD respondents reflects total accessible population in Metro Manila (hard-to-reach, low-prevalence group). Purposive sampling is appropriate for both.

Likert Scale — 5 Point

5 – Strongly Agree	*Excellently implemented*
4 – Agree	*Adequately implemented*
3 – Neutral	*Partially meets criteria*
2 – Disagree	*Insufficiently implemented*
1 – Strongly Disagree	*Poorly implemented*

Interpretation Ranges

4.21–5.00	Strongly Agree
3.41–4.20	Agree
2.61–3.40	Neutral
1.81–2.60	Disagree
1.00–1.80	Strongly Disagree

Statistical Formulas — Weighted Mean, SD, Overall WM

Weighted Mean (WM)

$$WM = \frac{(x_1 \times 5) + (x_2 \times 4) + (x_3 \times 3) + (x_4 \times 2) + (x_5 \times 1)}{n}$$

Where: x_1=Strongly Agree freq, x_2=Agree, x_3=Neutral, x_4=Disagree, x_5=Strongly Disagree; n=total respondents
Purpose: Single representative score per survey item for cross-group comparison.

Standard Deviation (SD)

$$SD = \sqrt{\frac{\sum_{i=1}^{k} f_i (w_i - WM)^2}{N}}$$

Low SD (<1.00) = high rater consensus; High SD (>1.00) = divergent opinions — interpret WM with caution.
Population formula used (each group = entire defined population, not a random sample).

Overall Weighted Mean (OWM)

Characteristic-level: average WM of all items under each ISO/IEC 25010 quality characteristic.
System-level: grand WM per respondent group (based only on assigned quality characteristics).
Passing threshold: WM ≥ 3.41 → 'Agree' = system satisfactorily meets the evaluated criterion.

Research Ethics

Data Sourcing

All government-produced public visual materials (DPWH, PAGASA, QC offices, barangays) are public domain under Philippine law. No private, proprietary, or personally identifiable data collected.

No Human Participant Research (Current Phase)

System evaluated entirely through mathematical metrics (ΔE, mAP50, CRR). No CVD individuals recruited, surveyed, or tested in the current cycle — noted as a starting point for future work.

Privacy by Design

Per-user Row Level Security (Supabase RLS) — users can only access their own images. 24-hour automatic deletion via scheduled Edge Function. Privacy policy disclosed at registration.

Person Exclusion Class

The exclude-person annotation class ensures no color enhancement is ever applied to humans in images. No recognizable individuals retained in the labeled dataset.

Not a Medical Device

ACRA is a digital visual aid — not a medical treatment, clinical intervention, or biological cure for CVD. This disclaimer applies to all versions and derivative works.

Research Summary

Problem	Color-coded public materials are inaccessible to ~300M CVD individuals; existing tools over-correct and destroy visual meaning.
Innovation	ACRA combines YOLOv8m semantic detection + Machado CVD simulation + CIEDE2000-bounded re-encoding — first documented semantic-aware CVD correction framework for Philippine public materials.
Architecture	IoT (Pi 5) → Cloud (FastAPI/Render) → Storage (Supabase) → Frontend (React/Vercel). Fully browser-accessible, privacy-preserving, scalable.
Dataset	1,274 → 3,312 images, 33,774 annotations across 5 semantic classes from real Philippine public materials.
Evaluation	ISO/IEC 25010 quality model; 3 respondent groups (5 IT experts + 5 optometrists + 10 Deutan CVD); thresholds: mAP50 ≥ 0.70, ΔE improvement > 15, CRR > 80%.
Ethics	Public domain data, no human participant testing in current phase, privacy-by-design (RLS + auto-delete), defined as digital visual aid not a medical device.

Thank You

Thank You